



■■ CHHATRAPATI SHAHU JI MAHARAJ UNIVERSITY, KANPUR
RESEARCH & DEVELOPMENT (R&D;) SECTION

NIVARAN-AI (■■■■■■■■-AI)

AI-Assisted Grievance Redressal & Workflow Automation System

A comprehensive technical design, architecture, and operational manual detailing the multi-tier academic grievance redressal platform, NLP classification pipeline, dynamic authority routing, in-app notification center, and Dean executive analytics.

Document Version:	v2.4.0 (Production Ready)
Target Entity:	Research Scholars, Students & University Administration
Core Stack:	FastAPI, React 18, PostgreSQL 14+, Scikit-Learn, SMTP
Automated Test Suite:	100% Passed (7 Test Suites, 32 Scenarios)
Release Date:	August 2026



1. Executive Summary & Vision

NIVARAN-AI is an enterprise-grade grievance redressal automation platform tailored for university research scholars and academic administrators at Chhatrapati Shahu Ji Maharaj University (CSJMU), Kanpur. Historically, grievance resolution in large academic institutions suffered from manual paper workflows, routing confusion between subject specialists and domain deans, lost attachments, and lack of accountability for inactive cases.

NIVARAN-AI resolves these bottlenecks by combining **Natural Language Processing (NLP)** with a **Multi-Tier Human-in-the-Loop Workflow**. The platform intelligently classifies grievance narratives, verifies them with University Managers, auto-dispatches cases to appropriate Assistant/Associate Deans, enables bidirectional document requests, provides real-time in-app & email notifications, and alerts authorities when cases remain inactive for 3 days.

Core Objective	System Capability	Impact
Automated Intake & NLP	TF-IDF + statistical classifier predicting categories with confidence scoring	Eliminates misrouting & categorizes text in <100ms
Multi-Tier Dynamic Routing	Dispatches between Subject Assistant Deans, Domain Associate Deans, & Fixed Officers	Zero confusion on officer assignment
Document Request Loop	Interactive pause/resume state machine with in-app document viewer	Eliminates repeated offline email ping-pong
3-Day Inactivity Reminders	Background jobs alert authorities if no action is recorded for ≥ 3 days	Prevents cases from stalling indefinitely
Dean Analytics Matrix	Live KPI dashboard aggregating officer workload, pending cases, & turnaround hours	Complete executive transparency for policy decisions

2. System Architecture & Layered Design

NIVARAN-AI is architected with a strict separation of concerns across presentation, API gateway, domain business services, machine learning inference, and data persistence layers.

Layer	Technologies	Responsibilities
Presentation Layer	React 18, Vite, Lucide Icons, Custom CSS Theme	Single Page Application (SPA), role-based dashboards, in-app notification bell, document modal viewer
API & Gateway Layer	FastAPI, Uvicorn ASGI, OAuth2 Bearer, Pydantic	RESTful routing, JWT authentication, request validation, CORS origin management
Business Service Layer	Python 3.10+, SQLAlchemy Services	State machine transitions, multi-tier escalation, document request lifecycle, reminder engine, email dispatch
AI / ML Engine	Scikit-Learn, TF-IDF Vectorizer, Multinomial NB	Text preprocessing, feature extraction, multi-class probability scoring, HITL audit logging
Persistence Layer	PostgreSQL 14+, SQLAlchemy 2.0, Alembic	19 relational tables, relational constraints, foreign keys, index optimization, 24 migration revisions
Communication Layer	SMTP, Jinja2 / HTML Templates, Polling	Responsive branded HTML emails for key events, unread in-app notification count badges



3. Administrative Hierarchy & RBAC

NIVARAN-AI enforces strict Role-Based Access Control (RBAC). 15 permission primitives govern every endpoint to ensure that scholars, operational managers, faculty officers, and executive deans operate strictly within their authority scope.

Role Name	Key Responsibilities & Scope	Data Isolation Scope
APPLICANT (Scholar)	Submits grievances, uploads receipts, responds to doc requests, monitors live timeline	Isolated strictly to own filed grievances & notifications
MANAGER (R&D;)	Verifies AI predictions, confirms/overrides categories, dispatches auto-routing, validates closure	University-wide intake & administrative closure queue
ASSISTANT DEAN	Assigned by scholar subject discipline; requests documents, directly resolves or forwards	Assigned subject grievances + active queue
ASSOCIATE DEAN	Receives forwarded cluster cases; resolves domain disputes; escalates policy issues to Dean	Assigned grievance cluster domain + active queue
DEAN (Executive R&D;)	Executive oversight, live workload performance matrix, final resolution of escalated disputes	Full university-wide visibility & analytics access

4. Grievance Redressal Lifecycle & Routing

Grievances transition deterministically through defined lifecycle states, ensuring comprehensive auditability and zero lost requests.

- **SUBMITTED:** Scholar submits title, description, priority, and optional supporting files. Background AI task queued.
- **AI_PROCESSING:** NLP engine computes predicted category, confidence score, and cluster mapping in the background.
- **PENDING_REVIEW:** Grievance displayed on Manager review queue for human-in-the-loop category confirmation.
- **ASSIGNED / IN_PROGRESS:** Dispatched to designated Assistant Dean. Officer investigates, requests files, or works on solution.
- **AWAITING_INFORMATION:** State paused while scholar uploads requested files. Restores automatically upon upload.
- **RESOLVED:** Authority submits resolution notes & proof. Scholar alerted via in-app notification and email.
- **CLOSED:** Manager verifies resolution quality, adds audit remarks, and formally locks the grievance.
- **REOPENED:** Scholar or Manager can reopen case if remedy is incomplete, returning grievance to active work.



Intelligent Multi-Tier Routing Scenarios:

Scenario	Trigger Domain	Automated Routing Path
Case 1: Subject Assistant Dean	Discipline-specific issues (Coursework, Supervisor, Lab Access)	Scholar → Manager → Subject Assistant Dean → Direct Resolution (No forwarding)
Case 2: Grievance Cluster	Cross-disciplinary domain issues (Hostel, Library, Exams)	Scholar → Manager → Assistant Dean → Mapped Associate Dean → Resolution
Case 3: Fixed Authority	Central administrative areas (Fellowship, Degree, Anti-Ragging)	Scholar → Manager → Assistant Dean → Dedicated Officer → Resolution
Escalation to Dean	Unresolved cluster disputes or major policy conflicts	Associate Dean → Dean (Prof. Namita Tiwari) → Final Resolution

5. AI Grievance Classification Engine

The NLP classification engine analyzes unstructured grievance narratives using statistical machine learning. It extracts unigram and bigram TF-IDF features across 5,000 vocabulary dimensions and computes multi-class category probability distributions.

Confidence Thresholding Policy:

- **High Confidence (≥ 0.75):** Highlighted with a green verification chip on the Manager dashboard, indicating strong alignment with historical category patterns.
- **Low / Medium Confidence (< 0.75):** Flagged with an amber warning, prompting the Manager to review the category dropdown and confirm or adjust the classification.

Human-in-the-Loop Audit Telemetry: Every prediction is logged in ai_processing_records (model name, latency in ms, confidence score, predicted category, final category, overridden flag, reviewer ID, timestamp), forming an active learning dataset for future model retraining.

6. Document Requests & 3-Day Inactivity Engine

Interactive Additional Document Request Workflow:

1. **Initiation:** Authority requests specific documents with required/optional flags and optional deadline.
2. **State Pause:** Grievance pauses in AWAITING_INFORMATION; scholar receives in-app alert and branded email.
3. **Fulfillment:** Scholar uploads files; status automatically restores to ASSIGNED / IN_PROGRESS with the same assigned officer.
4. **In-App Review:** Officer inspects documents in the embedded viewer: **Approve** or **Reject** (with re-upload loop).



3-Day Inactivity Reminder Engine:

- **Overdue Criteria:** Active grievance with $(\text{now} - \text{last_action_at}) \geq 3$ days and no reminder dispatched in current inactivity cycle.
- **Automated Actions:** Creates internal In-App notification & email alert for the assigned authority; updates `last_reminder_at`.
- **Duplicate Guard:** Consecutive runs create 0 duplicate notifications until new inactivity occurs.
- **Applicant Privacy:** The scholar is never sent internal inactivity reminders, preventing undue panic.

7. Dual-Channel Notifications & Email Delivery

NIVARAN-AI employs a dual-channel communication architecture combining a real-time In-App Notification Center with responsive, branded HTML email alerts.

Event Type	In-App Recipient	Email Dispatched?	Description
DOCUMENT_REQUESTED	Applicant	■ Applicant Email	Direct action link with requested file names & deadline
DOCUMENT_UPLOADED	Assigned Officer	■ No Email	Alerts officer that scholar submitted files
GRIEVANCE_FORWARDED	Target Officer, Applicant	■ No Email	Internal handoff alert with officer names
GRIEVANCE_RESOLVED	Applicant, Manager	■ Applicant Email	Resolution notes & optional proof link
GRIEVANCE_CLOSED	Applicant	■ Applicant Email	Formal administrative closure notice
REMINDER	Assigned Officer	■ (Internal only)	3-day inactivity alert for assigned officer

8. Dean Executive Analytics & Workload Matrix

The Dean Executive Dashboard provides real-time oversight of research scholar welfare across all 15 administrative authorities in the R&D; Section:

- **Executive KPI Cards:** Total grievances filed, currently pending cases, resolved volume, and university-wide average turnaround hours.
- **Authority Workload Matrix:** Officer-by-officer breakdown displaying Name, Role, Department (R&D;), Assigned Cases, Pending Bottlenecks, Resolved Volume, and Average Resolution Time.
- **Attention & Stalled Items API:** Flags grievances exceeding SLA limits or escalated to executive Dean level for immediate intervention.

9. Test Verification & Quality Assurance



NIVARAN-AI has undergone exhaustive end-to-end automated verification. All 7 test suites passed with 100% success rate:

Test Suite	Target Domain	Scenarios Verified	Status
<code>test_grievance_pipeline_e2e.py</code>	Core Routing Pipeline	4 E2E Scenarios (Cluster, Subject, Escalation, Proof)	■ 100% PASS
<code>test_email_notification_system.py</code>	Email Delivery & SMTP	9 Scenarios (Doc requests, resolve, close, SMTP fail)	■ 100% PASS
<code>test_notification_system.py</code>	In-App Notification Center	5 Scenarios (14 event types, unread count, mark-read)	■ 100% PASS
<code>test_reminder_feature.py</code>	3-Day Inactivity Engine	10 Criteria (Inactivity, duplicate guard, cycle reset)	■ 100% PASS
<code>test_document_request_system.py</code>	Additional Documents Loop	5 Scenarios (Request, upload, auto-restore, review)	■ 100% PASS
<code>test_dean_analytics.py</code>	Executive Dashboard	Workload matrix aggregation across all 15 officers	■ 100% PASS
<code>test_api_endpoints.py</code>	REST API Controllers	Intake, background task queue, document streaming	■ 100% PASS
<code>npm run build</code>	Frontend Production Bundle	65 modules transformed, 0 errors in 259ms	■ 100% PASS

Conclusion: NIVARAN-AI is fully stabilized, verified, documented, and prepared for seamless deployment at Chhatrapati Shahu Ji Maharaj University (CSJMU), Kanpur.